

Coder Survival — Deep Research Report

: 31 2026

: research-

: Telegram Mini App | Phaser 3.60 | Pixel-art idle/clicker

1. Executive Summary

: **Telegram Mini Apps** . MAU 1.44 (2024)
150–190 2025 tap-to-earn [63][68]. , —
« » , . Retention Mini Apps : **D1 = 15–20%**
26–33% [3][5]. , 60 Coder Survival — ,
 . ~**46%** web- [108], Telegram - - ,
21% [76]. **Phaser.CANVAS** pixelArt: true , :
Canvas **+30% FPS** WebGL [21], pixel art «1px
2px» [45]. Hamster Kombat , daily combo- **5** 300M
[65], Duolingo , **12** **streak >365** , Streak Wager 7-
retention **14%** [102]. : Coder Survival , idle-
(prestige, multipliers, offline income) UGC- — desktop-first UI
Telegram WebView.

2. Top 5

2.1. Canvas WebGL

: WebGL GPU- . : pixel art /
Phaser.CANVAS **30%** **FPS** [21]. — WebGL GPU (
Intel, Android) bottleneck - shader pipeline, Canvas 2D
 . pixel art . , WebGL Phaser 3
1px — 2px [45].

2.2. Telegram WebView CSS safe-area-inset-bottom

iPhone Face ID (X) env(safe-area-inset-bottom) home indicator.
CSS, Safari Chrome iOS, **Telegram Mini Apps** — 0
[80]. , UI (,) home indicator. :
window.visualViewport.height [78] padding ~34px iPhone
Face ID. Coder Survival desktop-first UI — ,
 .

2.3. Tap-to-earn , —

Notcoin , «tap-to-earn » [25]. Hamster Kombat
[68]. , — daily combo, daily cipher, streak bonuses, referral program
— . Hamster Kombat daily combo (3 = 5)
[65], daily cipher [92]. Coder Survival:
daily combo «daily stack» — 3 / , « » .

2.4. Duolingo **\$964M** ,
Streak — « ». Duolingo : 12 streak >365
, Streak Wager 7- retention **14%**, «Streak Freeze» «Weekend Amulet»
[102][103]. Coder Survival streak «Continuous Integration» — , CI
« ».

2.5. 21% **Telegram** — -
Magnetto, ~**21%** **Telegram** - [76]. Coder Survival —
, , . , Telegram Premium (15) ,
[71]. 25-34 **29.7%** [71] — IT- ,
in-app .

3. Telegram Mini Apps —

3.1. retention:
retention Telegram Mini Apps , . Monetag ,
D1 = 15–20%, D7 = 8–10% [3]. , 2024 **D1 =**
26–33%, D7 = ~8%, D30 < 3% [1][5]. Arcade- (clicker/tapper) D1 **22%**,
D28 ~**1%** [5]. : Mini Apps (, « »),
.

/	D1	D7	D30	
Telegram Mini Apps	15-20%	8-10%	/	Monetag [3]
()				
(25%)	26-28%	7-8%	< 3%	GameAnalytics [5]
(iOS)	31-33%	—	—	GameAnalytics [5]
(Android)	25-27%	—	—	GameAnalytics [5]
Idle/clicker (25%)	~ 42%	—	—	Mind Studios [14]
Hamster Kombat	—	—	—	300M [25]
()				

3.2. retention
Mini Apps idle- , **Daily rewards**
— , streak- . Liftoff, live events
adoption [11]. **Rewarded video ads** LTV **10–30%** -IAP [16],
interstitial ads
[14]. **Combo-** (Hamster Kombat) —
« », (). **Referral program** — Hamster Kombat 300M
[25]. **UGC + community:** Hamster Kombat
combo cipher [92] — , .

3.3. Telegram Desktop vs Mobile:
desktop/mobile Telegram — Coder Survival desktop-first .
desktop ~**46.35%** web- [108], Telegram . YouGov, **67%**

Telegram [101], **41** [71]. : **Telegram** — , , . desktop- (web-) Mini App — Telegram. **Coder Survival** , (), , ~**50%** **Telegram** **Mini Apps** [63], gaming- (-9%), **Tools/** Utilities (+15%) [68]. **Coder Survival** — « », « ».

3.4. **WebView**

Telegram Mini App **WebView** — . **iOS** : **WebView** **CSS safe-area-inset-bottom** [80], viewport height [82], iPhone 11/12 Pro Max **webview** [54]. **Android** : **WebView** - **TLS/Cloudflare** [58], **Android System WebView** [59]. (deep links) **Android**, **iOS** [62]. **Phaser-** : **WebGL**, **WebView** **GPU** — .

	iOS	Android	Coder Survival
safe-area-inset	[80]		
viewport	[82]		
	[54]		
WebView		[58]	
deep links	[62]		

3.5. **Mini Apps**

Telegram , **viral distribution:** — , , . **Hamster Combat** 0 300M [25]. **Telegram Ads:** **Telegram** **Mini Apps** **CPC \$0.01** **Tier-2/3** [68]. **TON / Stars** : **Telegram Stars** **TON** [10]. **Push-** : , **Mini App**, [10]. **Instant launch:** , **App** **Store review** — . **A/B-** .

4. / / —

4.1. **Telegram Mini Apps**

Coder Survival **Telegram** — **Mini App** / . — **Hamster Combat** (crypto exchange CEO), **Catizen** (), **TapSwap** () — [24][26]. . **Hamster Combat** **GameDev Heroes** — , [66]. : **Office Tab**, **HR-** , **Development** **Marketing** , **combo rewards** . **Hamster Combat** , « - » .

4.2. idle/clicker

incremental/idle games [20]. **AdVenture Capitalist** — : prestige Angel Investors (-) [112], - , exponential cost scaling, multiple planets/ . **Cookie Clicker** — : + + prestige, [18]. **Idle Miner Tycoon** — digging building up [14]. : **core loop = tap → earn → upgrade → tap faster → prestige → repeat**.

Coder Survival : tap = , earn = / , upgrade = / , prestige = / . — **exponential cost scaling**: cost = base × (growth_rate ^ level) [119]. , = 10, growth_rate = 1.15, level 1 10, level 5 — 20, level 20 — 163, level 100 — 1,174,000. , prestige.

4.3. : idle-

idle- : **60–70%** , **30–40% IAP** [14]. Rewarded video ads — [13]. Interstitial ads 2 , « » IAP [14]. Telegram Mini Apps TON Stars TON — [10]. Coder Survival IAP: Streak Freeze, offline income, premium (5 = 5 +), « », Battle Pass. : **onetization must not harm retention** — [14].

4.4. ,

— Coder Survival. Hamster Kombat «hamster CEO» [25]. Notcoin — , Coder Survival : (,), , «negative events», Stack Overflow , - . UGC- () **core loop**: prestige/sprint « », Telegram. viral loop — = .

4.5. streak / daily bonus —

Duolingo, Hamster Kombat . **Increasing rewards** — streak, (diminishing returns). **Milestone celebration** — 7, 30, 50, 100, 365 [50]. **Grace mechanisms** — Streak Freeze, Weekend Amulet, «soft decay» hard reset [37][41]. **Social streak** — 1/3 Duolingo Friend Streak [102]. Coder Survival streak «Continuous Integration»: — , = «failed build». Streak Saver () = «rollback» CI. — « » , « ».

	Duolingo	Hamster Kombat	Coder Survival ()
Core streak mechanic	[103]	Daily combo cards [65]	CI commits ()
Grace mechanism	Streak Freeze [37]		Streak Saver (rollback)
Milestone rewards	Confetti animation [50]	5M coins combo	«Release» celebration
Social element	Friend Streak [102]		Team/code review streak
Monetization	Streak Wager (+14% retention) [102]	—	Streak insurance (IAP)

5. Phaser 3.60 — pixel-art -

5.1. - pixel-art

- — Phaser- . 100 (100 HTTP- , 100 draw calls) [35]. pixel art 5 (idle, typing, bug, coffee, rage) : (coder-atlas.png + coder-atlas.json) , UI- , . (parallax). **Lazy loading** / — [64]. : TexturePacker (, \$39) [35] Atlaspack Pixel Tools (open source, Go) [12].

5.2. Phaser.CANVAS

Phaser.CANVAS Phaser.WEBGL : (1) pixel art WebGL — batching [42]. (2) Canvas **+30% FPS** [21]. (3) Canvas 1px — WebGL 2px - [45]. (4) Canvas — GPU texture memory. (5) Pixel-perfect rendering: pixelArt: true nearest-neighbor filtering, [43]. Canvas: tinting, shader effects, 9-slice sprites [45]. **Coder Survival** — pixel art .

5.3. vs :

Phaser : (,) (Graphics API). : , UI- , , , . : (progress bars, health bars), , debug visualization, procedural backgrounds. **Coder Survival** : (, ,) — / ; progress bars, energy bar, depression meter — Graphics API (,).

5.4. WebView (~150MB) heap

WebView . **Android:** heap size , 16–64MB , android:largeHeap="true" () [33]. **iOS:** RAM ~160MB , ~2GB , WebView [33]. **Phaser-** **1.3GB RAM** WebView iPad, kills OS [17]. **Coder Survival** : 5 , - , UI- ~50–80MB. **Phaser 3** **memory leak** — [70]. : textures.remove(key) lazy loading [74].

Texture atlas	(draw calls 10–100×)
Object pooling	(GC-)
Lazy loading	()
Image compression (TinyPNG)	
Canvas WebGL	(+30% FPS)
Integer positions	

6. combo system -

6.1. combo-

Hamster Kombat Codex Survival. **Daily Combo** — ,
 3 (Markets, PR & Team, Legal, Specials) 5 [65]. Combo
 12:00 GMT. : , , [92]. **Daily**
Cipher — : = , = 2 , = 1 [90].
 Codex Survival : Daily Combo = «daily stack» (3), Daily
 Cipher = «daily commit message» (/).

6.2. (x2, x3, x5):

Combo- **Coder Survival** (x1.5/2/3/5 5/10/20/50) , :
x2 — « », , satisfaction ^[118]. **x3** — , .
x5 — , , . : —
 , , , , ^[118]. , . :
(x1) → (x1.5) → (x2) → (x3) → (x5). x5 — **screen shake** + **confetti**
 { } ; !.

6.3. :

visual feedback combo
+1 LOC, — x2 COMBO! [118]. **Progress bar** —
; , . **Screen effects** — screen shake
x3, x5; chromatic aberration 100ms. **Avatar reaction** — combo:
idle (x1), typing (x2), «in the zone» (x3), «god mode» (x5). **Sound design** —
x5 «level up» jingle.

6.4. combo retention

Combo- retention . **Variable reinforcement** —
 () « » ^[41]. **Zeigarnik effect** — combo (49/50)
 « », ^[41]. **Loss aversion** — combo = ;
 (Streak Saver) ^[102]. : combo
 rewarded ads « » « ». Hamster Kombat combo, —
 — ^[65].

7. Gap Analysis — Coder Survival

7.1. :

Phaser.CANVAS, pixelArt: true		pixel art, +30% FPS
Combo (x1.5/2/3/5 5/10/20/50)		,
5 (idle, typing, bug, coffee, rage)		,

Table 5 – continued

Streak Saver ()	✓	retention,	CI-
UGC (+)	✓	,	prestige
Battle Pass catch-up	✓	,	tiers
Desktop-First UI (60%+ Desktop)	✓	Telegram [76]	

7.2. ()

Prestige / soft reset	idle- ; AdVenture Capitalist prestige [112]. replayability.	GDC talks [15]
Daily combo (3 =)	Hamster Combat: 5M , [65]	Hamster Combat [92]
Offline income	Idle- FOMO: [14]	Mind Studios [14]
Grace period / Streak Freeze	Duolingo: 12M+ streak >365; freeze –burnout churn [102]	Duolingo [37]
Rewarded video ads	LTV 10–30%; idle- [16]	Adjust [16]
Telegram WebView viewport fixes	iOS: safe-area-inset-bottom , height [80][82]	GitHub issues [54]
Lazy loading	WebView ~150MB; Phaser memory leak [70]	Phaser issues [17]
Referral / viral loop	Hamster Combat 300M [25]	KuCoin [26]
Exponential cost balancing formula		Medvešček [119]
Object pooling	day 3 [119]	
Push-	GC- 60 FPS [64] Roman (TMA): « boost retention» [10]	Generalist Programmer [64] Monetag [10]
Dual currency (Stars + TON)	« –enable both» [10]	Monetag [10]

7.3.

P0 (Must have)	Prestige / soft reset	–	idle
P0	WebView viewport fixes (iOS)	iOS	
P0	Exponential cost formula		

Table 7 – continued

P1 (Should have)	Daily combo	D1/D7
P1	Offline income +	Core idle
P1	Rewarded video ads	
P1	Streak Freeze / grace period	churn
P2 (Nice to have)	Referral program	Viral growth
P2	Push-	Re-engagement
P2	Object pooling	Performance

8.

8.1. P0 — Must have ()

1. prestige / soft reset (« »)

Prestige — «nice to have», . Coders Survival 3–5
 : « » (soft reset), permanent — « »
 (XP), . : angels = sqrt(total_L0C / 10) — ,
 . prestige 1–2 (50–150 XP), x2–x4 [112]. prestige
 XP . UI: « » , .

2. exponential cost balancing

$(\text{growth_rate} \wedge \text{level})^{[119]}$. $(\text{TypeScript} = 10 \text{ LOC}, \text{growth_rate} = 1.15, \text{Level} = 10, \text{level} = 5 = 20, \text{level} = 20 = 163, \text{level} = 100 = 1.17\text{M})$. $\text{cost} = \text{base_cost} \times \text{multipliers}$, $\text{income} = \text{base_income} \times \text{level} \times \text{prestige}^{[113]}$.

3. WebView viewport iOS

index.html:

```
<meta name="viewport" content="width=device-width,initial-scale=1,maximum-scale=1,user-scalable=no,shrink-to-fit=no,viewport-fit=cover">
```

JavaScript:

```
if (window.visualViewport) {
  window.visualViewport.addEventListener('resize', () => {
    document.body.style.height = window.visualViewport.height + 'px';
  });
}

window.addEventListener('scroll', () => {
  if (window.scrollY > 0) window.scrollTo(0, 0);
});
```

CSS: `html, body { overflow: hidden; overscroll-behavior: none; }`. padding 34px
iPhone Face ID [78].

8.2. P1 — Should have ()

4. Daily Combo («Daily Stack»)

12:00 GMT 3 / . 3 — (,
1 x2 income). UGC (). Hamster
Kombat [65], .

5. offline income

« » offline, 4 («backlog»). accumulation
— FOMO « » [14]. UI: « X !»
“Claim”.

6. rewarded video ads

rewarded ads : « 2x income 5 », « Streak Freeze»,
« +1 combo». ads 3 — D1 retention [14]. Telegram Mini Apps
AppLixir [2].

7. Streak Freeze streak

Streak Freeze. — IAP. «Weekend Amulet» — streak
[37]. churn - .

8.3. P2 — Nice to have ()

8. referral

« — ». Hamster Kombat 300M [25].
Telegram — . : +10% income 24 (stackable).

9. push-

, Mini App, : « CI 2 !», «Daily Combo », « 1000
offline». Roman (TMA) : «enable push notifications for bots that launch your apps
— they nicely boost retention rates» [10].

10. : lazy loading + object pooling

- (title, gameplay, prestige), [74]. object pooling
(floating text, confetti) — GC- [64]. footprint: < 80MB RAM.

. 2026.